

“Never ignore a coincidence. Unless you’re busy, then always ignore a coincidence.” 11th Doctor

IE 4516 Quality Assurance: Recitation

THE DESTRUCTION OF REALITY ITSELF



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The Master Master Plan

The Master is back with a plan to create the **ULTIMATE CHAOS** on planet Earth and destroy Dr. Who!



For this, they decided to create an **Experiment** to see who are the **Doctor Who's most powerful Villains**. They want to create the most chaotic and disruptive situation that Earth has ever seen.

They are going to go back and forth in Time to measure the Chaotic Index © when some of the villains' attack.

The Chaotic Index © is a patented estimate of the Financial, Life, and Operational Disruption that Earth is facing at a given moment.

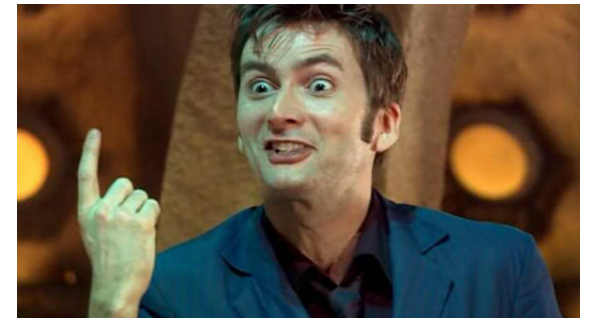
Factors that affect the Chaos Index

The controlled factor affecting the Chaos Index © is the attack of different species on Earth.

Each of the attackers might have a significant effect on the Chaos Index. Your goal is to find which villains from the 6 different aliens have a significant individual effect on the result.

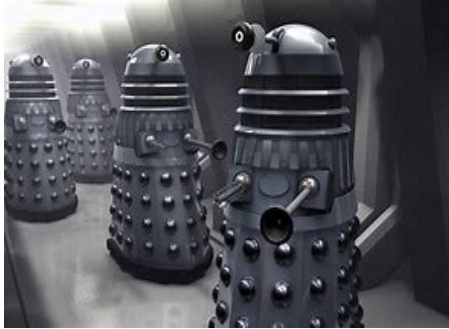
The Master provided the observations he collected from their Experiment. It is uploaded on the Super Secret Location on Canvas under the name **Index.xlsx**

“First thing’s first,
but not necessarily
in that order.””



The Villains

DALEKS



“EXTERMINATE”

THE EMPTY CHILD



“Are you my Mummy?”

THE SILENCE



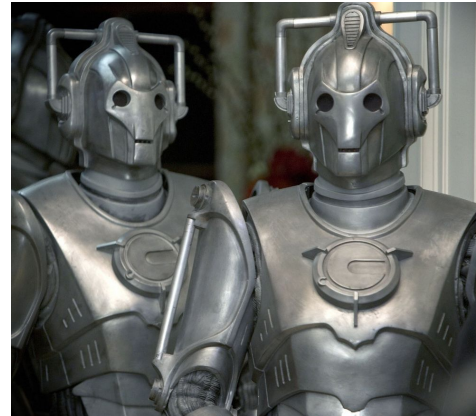
"Silence must fall when the Question is asked"

AUTONS



You've got a very distorted sense of humor, haven't you?

CYBERMEN



“You will not be harmed. You will be upgraded”

ZYGONS



“20 million Zygons have been allowed to take human form”

Part I: 1-Factor ANOVA Experiment

Goal 1: Increase the chaos on Earth recommending the instigation of one of the villains of Doctor Who series.

Response: Chaotic Index ©

Factor: Villain attack

Levels (6): Type of Villain

Autons,
Daleks,
Cybermen

The Silence,
Zygons and
The Empty Child



“Do what I do: Hold tight and pretend it’s a plan!”



Part I: Excel worksheet

START WORKING ON THE DATA

Treatments	Observations						Sums	Averages	SS	SUMS ²
Dalek	71	61	90	82	68	61				
Cybermen	87	80	61	50	89	83				
Zygons	59	51	85	78	52	31				
Autons	45	83	66	23	46	63				
The-Silence	28	58	52	53	19	26				
Empty-Child	90	37	52	63	23	53				
Totals										

	SS	dof	MS	F ₀	p-value
Treatment					
Error					
Total					

n = 6
a = 6 (# of villains = levels of Factor Villains)
N = ?

Part I: 1-Factor ANOVA Experiment

We will use **Excel** to analyze this simple problem.

You can use **Minitab** to make graphs and verify your calculations.

Part I: Remember how to calculate

Typical Data for a Single-Factor Experiment						
Treatment	Observations				Totals	Averages
1	y_{11}	y_{12}	...	y_{1n}	$y_{1\cdot}$	$\bar{y}_{1\cdot}$
2	y_{21}	y_{22}	...	y_{2n}	$y_{2\cdot}$	$\bar{y}_{2\cdot}$
.	.	.	...	.	.	.
.	.	.	...	.	.	.
.	.	.	...	.	.	.
a	y_{a1}	y_{a2}	...	y_{an}	$y_{a\cdot}$	$\bar{y}_{a\cdot}$
					$\bar{y}$	$\bar{y}_{..}$

Total of all n observations of level 1

Average of all n observations of level 1

Average of the average

Total of the total

Part I: Remember how to calculate

■ **TABLE 4.6**

The Analysis of Variance for a Single-Factor Experiment

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square	F_0
Treatments	$SS_{\text{Treatments}}$	$a - 1$	$MS_{\text{Treatments}}$	$\frac{MS_{\text{Treatments}}}{MS_E}$
Error	SS_E	$a(n - 1)$	MS_E	
Total	SS_T	$an - 1$		

Part I: WHO?

ARE THE ATTACKES **STATISTICALLY SIGNIFICANT** IN AFFECTING THE CHAOS INDEX©?

WHICH VILLAIN GROUP IS EFFECTIVE IN CREATING CHAOS?

OR

WHICH VILLAIN ATTACK MAKES A **STATISTICALLY SIGNIFICANT DIFFERENCE** IN THE CHAOS INDEX©?

Part II: CHAOTIC WAR

Now, the Master realized that they could create further Chaos if he engaged in **multiple attacks at the same time!**
They want to know how one attack from one Villain group might affect the other Villain attack.

They also want to know what is the
best attack **strategy!!**

What is the **best combination**
of **high and low levels of attack among Villains**
that would make the **most impact** on Earth?



**“Use your enemies’ power
against them.” 12th Doctor**

Part II: Minitab calculations

The Master provided the observations he collected from their Experiment.

It was uploaded on the Super Secret Location on Canvas under the name **Chaotic-War.xlsx**

Paste into Minitab and select DOE> >Factorial > Analyze Factorial Design> (Yes)

Select only appropriate factors

Evaluate only 3-order interactions

Part II: Minitab calculations

Define Custom Factorial Design

C1 Y=Adhesion
C2 Agent
C3 Cleanser
C4 Mordant

Factors:
Agent-Mordant

☒ 2-level factorial
☐ 2-level split-plot (hard-to-change factors)
☐ General full factorial

Factors from above that are hard to

Select

Low/High...

Designs...

Help

OK

Cancel

C3	C4						
Cleanser	Mordant						
-1	-1						
-1	-1						
1	-1						
-	-						

Analyze Factorial Design

C1 Y=Adhesion

Responses:
Y=Adhesion

Terms...

Covariates...

Options...

Stepwise...

Graphs...

Results...

Storage...

Select

Help

OK

Cancel

Part II: Minitab calculations

Should look familiar

Analysis of Variance



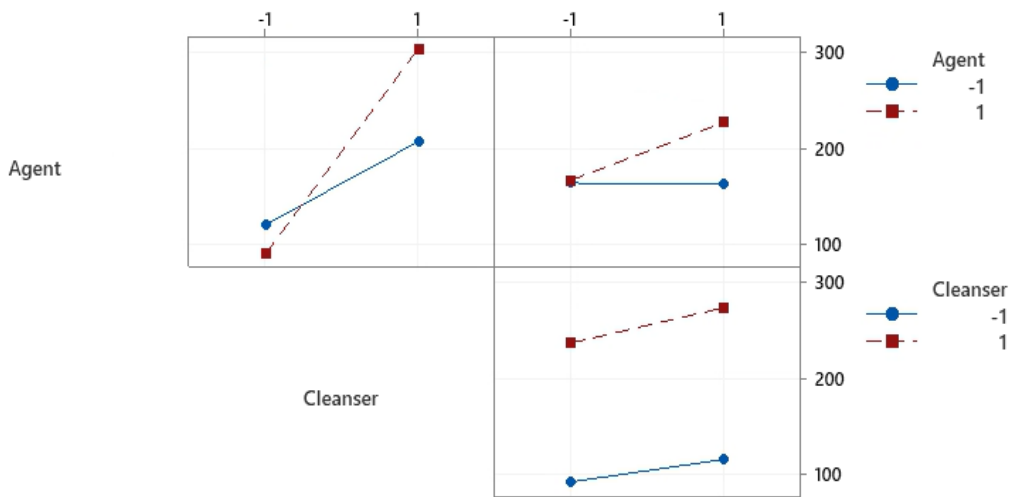
Source	DF	Adj SS	Adj MS	F-Value	P-Value
Model	7	128686	18383.8	229.62	0.000
Linear	3	98777	32925.7	411.25	0.000
Agent	1	4456	4455.6	55.65	0.000
Cleanser	1	90752	90751.6	1133.51	0.000
Mordant	1	3570	3570.1	44.59	0.000
2-Way Interactions	3	20059	6686.2	83.51	0.000
Agent*Cleanser	1	16193	16192.6	202.25	0.000
Agent*Mordant	1	3691	3690.6	46.10	0.000
Cleanser*Mordant	1	176	175.6	2.19	0.177
3-Way Interactions	1	9851	9850.6	123.04	0.000
Agent*Cleanser*Mordant	1	9851	9850.6	123.04	0.000
Error	8	641	80.1		
Total	15	129327			

This one is not the real one yet
NO SPOILERS

Part II: Minitab calculations

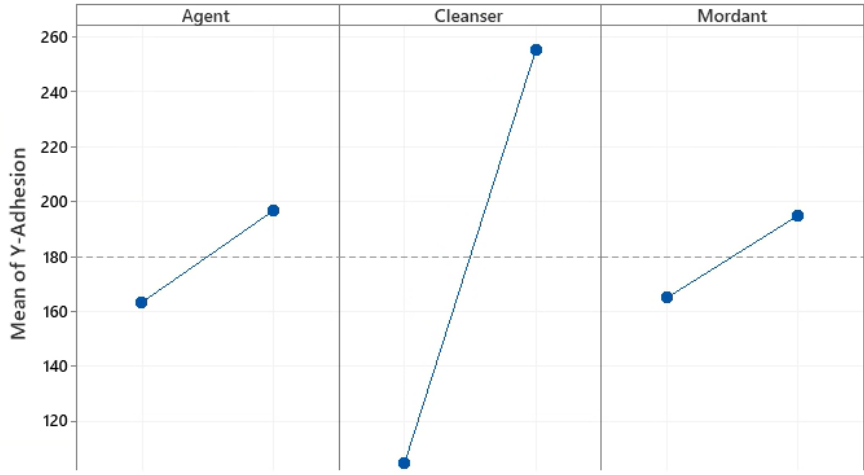
Factorial Plots from Factorial menu
Box plots from Graph Menu

Interaction Plot for Y-Adhesion
Data Means

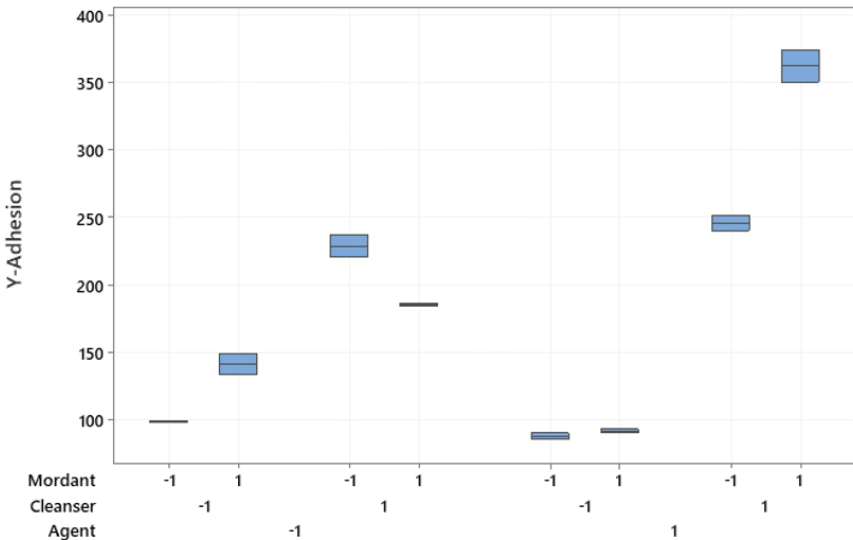


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Main Effects Plot for Y-Adhesion
Fitted Means



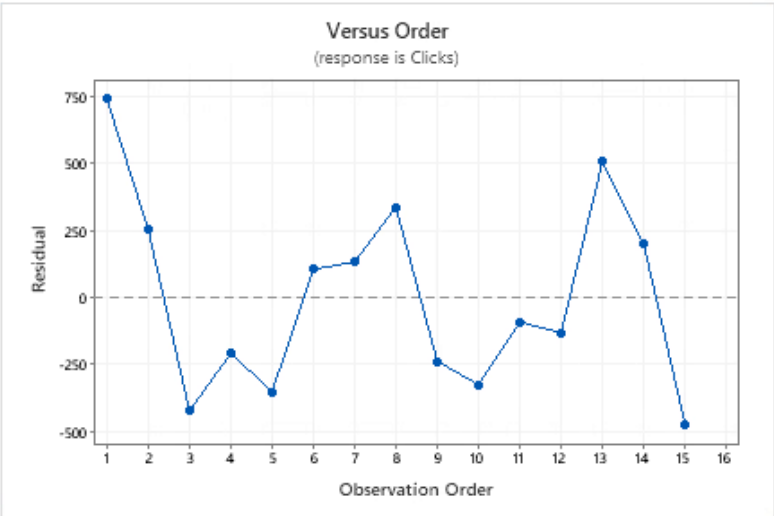
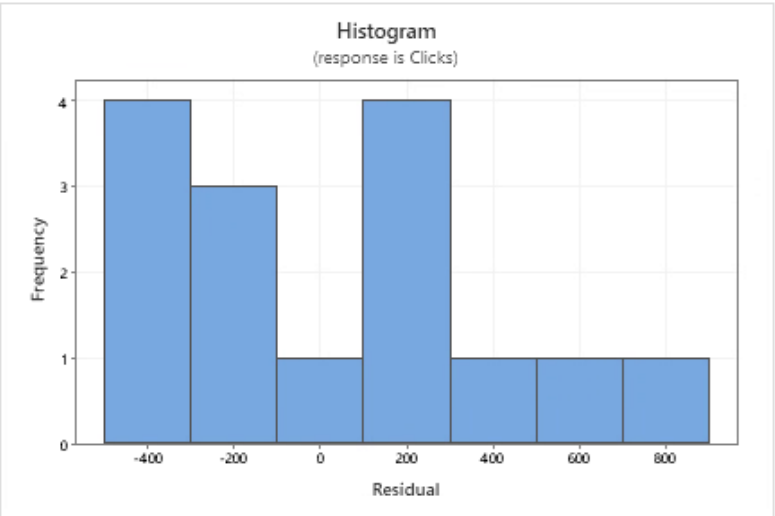
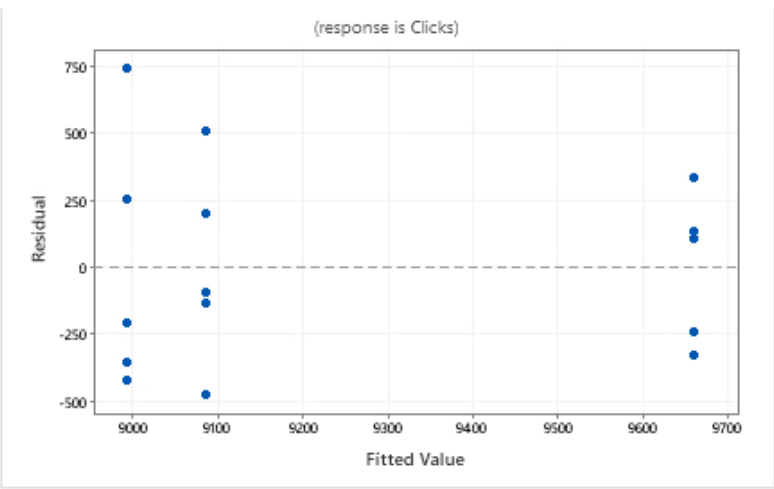
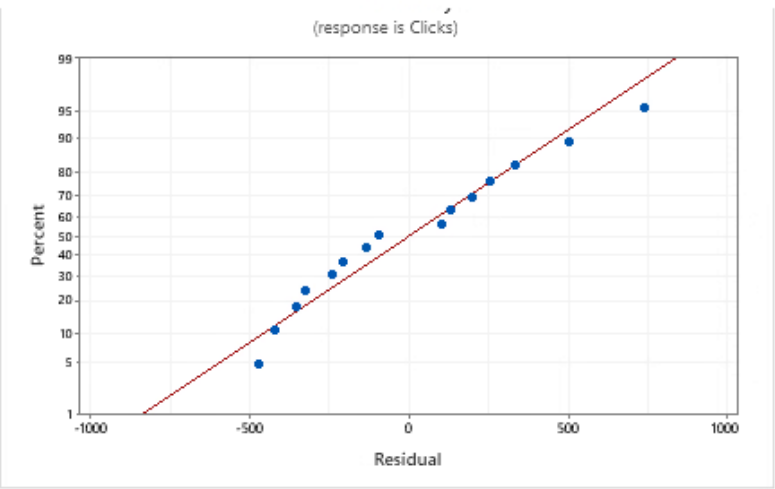
Boxplot of Y-Adhesion



Part II: Minitab calculations

Residuals from Graphs option,
ANOVA menu

General Linear Model: Clicks versus Treatment



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NO SPOILERS

Part II: Minitab calculations

Do Linear ANOVA – are settings significant?

In the Model tab, select interactions (all, through **order 3**) – are they significant?

In the Graph tab, look at residuals – is the random noise in the system normal?

Under Stat > ANOVA look at main effect and interaction plots – learn anything?

Box Plots? Anything else?

Part II: Conclusions and Recommendations

What would you recommend to The Master?

Who should he hire?

At what level each Alien group should engage in this Chaotic War to create the most effect?

Justify your answers with the results of the Experiment/Graphs!

Credit for Today

UPLOAD on Canvas with team names - one per team

Basic Credit – upload the Excel sheet + Word/PDF file with Part II graphs/tables

Extra Credit – upload your team's conclusions from Part I and II and recommendations (Word/PDF file)

Variable based on how much you did with it and how much sense your conclusions make!

Remember: The Master is one of the smartest and more powerful person on the whole Spacetime.

Don't try to cross them! ;)

I see you!

